

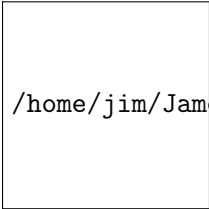
Section 3.6: Conditional Probability

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

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Conditional Probability

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$$p(A | B)$$

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Product Rule

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Probability of A if B has occurred.

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$$p(A \cap B) = p(A | B) \cdot p(B)$$

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$$p(\text{heart and heart are dealt}) =$$

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Example 3.6.4.

IF two cards are dealt from a full deck, find the probability that both are hearts.

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Tree Diagram

Using a Tree Diagram:

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Example 3.6.5.

If two cards are dealt from a full deck, find the probability that both are hearts.

Using a Tree Diagram:

Problem 3.6.1. Find the probability using a tree diagram.

Of two cards dealt from a full deck exactly one is a spade.

Problem 3.6.2. Find the probability using a tree diagram.

Of three cards dealt from a full deck exactly one is a spade.

Problem 3.6.3. Use the provided data to complete the following

Age	Obama	McCain	Other/NA
18-29	11.9%	5.8%	0.4%
30-44	15.1%	13.3%	0.6%
45-64	18.5%	18.1%	0.4%
65+	7.2%	8.5%	0.3%

Problems

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